This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Scaler
Manufacturer: Extron
Firmware Version: N/A
Model(s): IN1606, IN1608, IN1608 MA, IN1608 SA, IN1608 IPCP, IN1608 HDBT, IN1608 MA 70, IN1608 MA 70 HDBT, IN1608 SA HDBT, IN1608 IPCP SA, IN1608 IPCP SA HDBT, IN1608 IPCP MA 70, IN1608 IPCP MA 70 HDBT

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
3.10.0000-b007	2.6.0

Version History

Module Version	Date	Notes
1_7_0_0	8/31/2020	Changed driver to use SSH protocol. Added commands: • Mic Line Input Gain • Mic Line Input Mute Renamed commands: • Bass → Group Bass
1_6_0_0	3/27/2019	Fixed status for Ducking and VideoMute commands. • Renamed commands: ○ AmplifiedOutputMic → AmplifiedOutputMicLine ○ AnalogOutputMic → AnalogOutputMicLine ○ Digital Output Mic → Digital Output Mic Line ○ SwitchingEffect → SwitchTransition

		○ VariableAnalogOutputMic → VariableAnalogOutputMicLine • Renamed values: ○ Include Mic → Include and Exclude Mic → Exclude for AmplifiedOutputMicLine, AnalogOutputMicLine, DigitalOutputMicLine, and VariableAnalogOutputMicLine commands ○ L+R (Mono) → Dual Mono Program and Stereo → Stereo Program for AmplifiedOutputProgram command ○ Left → Left Program and L+R (Mono) → L+R Program (Mono) for AnalogOutput1Program command ○ Right → Right Program and L+R (Mono) → L+R Program (Mono) for AnalogOutput2Program command ○ Dual Mono → Dual Mono Program and Stereo → Stereo Program for DigitalOutputProgram and VariableAnalogOutputProgram commands ○ Fade → Fade Through Black for SwitchingTransition command ○ None → Off and Audio Test Pattern → Audio Test for TestPattern command • Renamed command parameters: ○ Mic → Mic/Line for Ducking command • Renamed command parameter values: ○ Digital → LPCM-2Ch for Type parameter for Input Gain Command
1_5_1_0	7/26/2018	Added Front Panel Volume Knob Control notes.
1_5_0_0	4/6/2018	Added all supported resolution for Output Resolution. Added TestPattern command. Added HDCPInputStatus and Temperature status. Combined Output1Mic and Output2Mic into AnalogOutputMic command. Renamed commands: • SAMAAmpProgram → AmplifiedOutputProgram • SAMAAmpMic → AmplifiedOutputMic • Output1Program → AnalogOutput1Program • Output2Program → AnalogOutput2Program • AudioInputType → AudioFormat • AudioOutputMute → AudioMute • HDMIABDTPCProgram → DigitalOutputProgram • HDMIABDPTCMic → DigitalOutputMic • MicMute → GroupMicMute • MicVolume → GroupMicVolume • OutputVolume → GroupOutputVolume

		• ProgramMute → GroupProgramMute • ProgramVolume → GroupProgramVolume • InputHDCPAuthorization → HDCPInputAuthorization • InputFormat → InputSignalType • AudioInputGain → InputGain • GainControl → OutputGain • HDMIOutputFormat → OutputFormat • VariableLRProgram → VariableAnalogOutputProgram • VariableLRMic → VariableAnalogOutputMic • VideoOutputMute → VideoMute Renamed Statuses: • InputStatus → InputSignalStatus • ScreenSaver → ScreenSaverStatus Renamed values for the following commands: • AmplifiedOutputProgram • AmplifiedOutputMic • AnalogOutput1Program • AnalogOutput2Program • AspectRatio • AudioInputType • AutoSwitchMode • GlobalVideoMute • DigitalOutputProgram • HDCPOutputStatus • Input • InputSignalStatus • VariableAnalogOutputProgram • VariableAnalogOutputMute • VideoOutputMute
1_4_4_2	9/29/2017	Update module to Rev. B
1_4_4_1	2/21/2017	Fixed AudioInputGain command. Updated password logic. Updated module notes.
1_4_4_0	8/26/2016	Updated Module.
1_0_0_1	2/26/2016	Fixed update issue.
1_0_0_0	1/21/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  `Example: InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  `Example: InterfaceName.connectionCounter = 5`

- If login credentials are required, devicePassword and deviceUsername must be set accordingly. deviceUsername default value is admin respectively.
  `Example: InterfaceName.deviceUsername = 'extron'`

- InputFormat command will not display status if the input selected does not have an active signal.

- Audio breakaway is not allowed to an input configured for any digital audio format.

- Video breakaway is not allowed from an input configured for any digital audio format.

Supported Classes and Examples

SerialClass

```
    InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='IN1606')
```

SerialOverEthernetClass

```
    InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='IN1606')
```

SSHClass

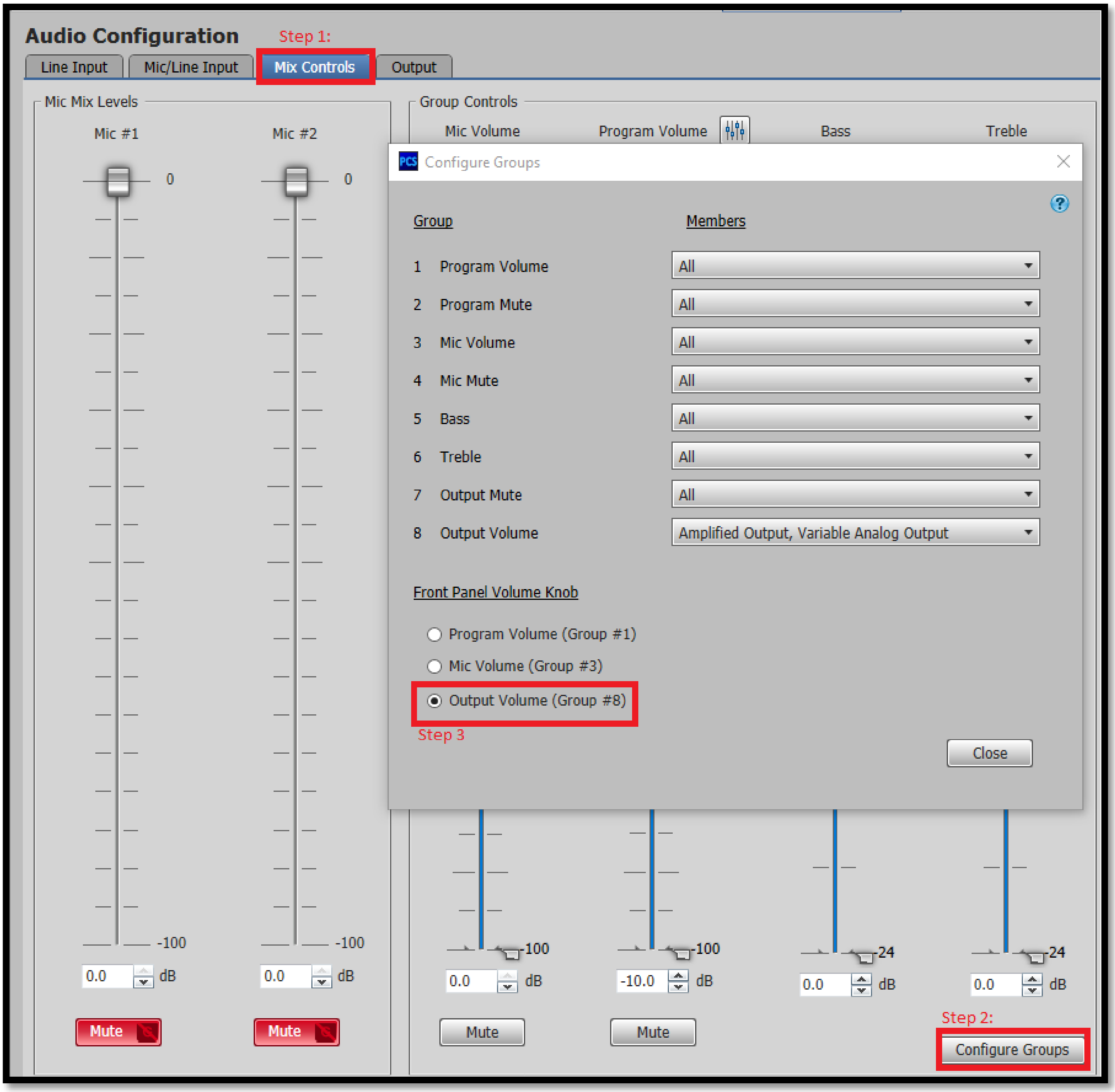
```
#Password Required
    InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', 'extron'), Model='IN1606')

#No Password Required
    InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', ''), Model='IN1606')
```

Front Panel Volume Knob Control:

To control the Front Panel Volume Knob of the IN1608 with this module, it must be properly configured in PCS.

Under Audio Config, go to Mix Controls and select Configure Groups. Under Configure Groups, configure the Front Panel Volume Knob to Output Volume (Group #8). When using the module, select the Group Output Volume command to control the Front Panel Volume Knob.

Note: Members of Output Volume (Group #8) can be adjusted to fit application needs.



Control Commands

Format with Qualifier:
```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:
```
InterfaceName.Set(Command, Value)
```

Command **AmplifiedOutputMicLine** [1]	Value 'Include'	Value 'Exclude'	
`# AmplifiedOutputMicLine example` `InterfaceName.Set('AmplifiedOutputMicLine', 'Include')`			
Command **AmplifiedOutputProgram** [1]	Value 'Dual Mono Program'	Value 'No Program'	Value 'Stereo Program' [3]
`# AmplifiedOutputProgram example` `InterfaceName.Set('AmplifiedOutputProgram', 'Dual Mono Program')`			
Command **AnalogOutput1Program**	Value 'Left Program'	Value 'L+R Program (Mono)'	Value 'No Program'
`# AnalogOutput1Program example` `InterfaceName.Set('AnalogOutput1Program', 'Left Program')`			
Command **AnalogOutput2Program**	Value 'Right Program'	Value 'L+R Program (Mono)'	Value 'No Program'
`# AnalogOutput2Program example` `InterfaceName.Set('AnalogOutput2Program', 'Right Program')`			
Command **AnalogOutputMicLine**	Value 'Include'	Value 'Exclude'	
Qualifier Key **'Output'**	Qualifier Value '1' – '2'		
`# AnalogOutputMicLine example` `InterfaceName.Set('AnalogOutputMicLine', 'Include', {'Output': '1'})`			
Command **AspectRatio**	Value 'Fill'	Value 'Follow'	
Qualifier Key **'Input'**	Qualifier Value '1' – '6' [4]	Qualifier Value '1' – '8' [5]	
`# AspectRatio example` `InterfaceName.Set('AspectRatio', 'Fill', {'Input': '1'})`			
Command **AudioFormat**	Value 'Analog' 'LPCM-2Ch Auto'	Value 'LPCM-2Ch' 'Multi-Ch Auto'	Value 'Multi-Ch' None
Qualifier Key **'Input'**	Qualifier Value '1' – '6' [4]	Qualifier Value '1' – '8' [5]	
`# AudioFormat example` `InterfaceName.Set('AudioFormat', 'Analog', {'Input': '1'})`			
Command **AudioMute**	Value 'On'	Value 'Off'	
Qualifier Key **'Output'**	Qualifier Value 'Output 1' 'Right Variable Output' 'Left Amplified Output' [3]	Qualifier Value 'Output 2' 'Left Digital Output' 'Right Amplified Output' [3]	Qualifier Value 'Left Variable Output' 'Right Digital Output' 'Amplified Output' [2]
`# AudioMute example` `InterfaceName.Set('AudioMute', 'On', {'Output': 'Output 1'})`			

Command	Value	Value	Value
AutoImage	'Execute'	'Execute and Fill'	'Execute and Follow'

```
# AutoImage example
InterfaceName.Set('AutoImage', 'Execute')
```

Command	Value	Value	Value
AutoSwitchMode	'Off'	'Highest Active Input'	'Lowest Active Input'

```
# AutoSwitchMode example
InterfaceName.Set('AutoSwitchMode', 'Off')
```

Command	Value	Value
DigitalOutputMicLine	'Include'	'Exclude'

```
# DigitalOutputMicLine example
InterfaceName.Set('DigitalOutputMicLine', 'Include')
```

Command	Value	Value	Value
DigitalOutputProgram	'Stereo Program'	'Dual Mono Program'	'No Program'

```
# DigitalOutputProgram example
InterfaceName.Set('DigitalOutputProgram', 'Stereo Program')
```

Command	Value	Value
Ducking	'Enable'	'Disable'
Qualifier Key	Qualifier Value	
'Mic/Line'	'1' – '2'	

```
# Ducking example
InterfaceName.Set('Ducking', 'Enable', {'Mic/Line': '1'})
```

Command	Value	Value	Value
ExecutiveMode [8]	'Off'	'Mode 1'	'Mode 2'

```
# ExecutiveMode example
InterfaceName.Set('ExecutiveMode', 'Off')
```

Command	Value	Value
Freeze	'On'	'Off'

```
# Freeze example
InterfaceName.Set('Freeze', 'On')
```

Command	Value	Value	Value
GlobalVideoMute	'On'	'On with Sync'	'Off'

```
# GlobalVideoMute example
InterfaceName.Set('GlobalVideoMute', 'On')
```

Command	Value
GroupBass	-24 to 12 in steps of 0.1

```
# GroupBass example
InterfaceName.Set('GroupBass', 12)
```

Command	Value	Value
GroupMicMute	'On'	'Off'

```
# GroupMicMute example
InterfaceName.Set('GroupMicMute', 'On')
```

Command	Value
GroupMicVolume	-100 to 0 in steps of 0.1

```
# GroupMicVolume example
InterfaceName.Set('GroupMicVolume', 0)
```

Command	Value	Value
GroupOutputMute	'On'	'Off'

```
# GroupOutputMute example
InterfaceName.Set('GroupOutputMute', 'On')
```

Command	Value
GroupOutputVolume [10]	-100 to 0 in steps of 0.1

```
# GroupOutputVolume example
InterfaceName.Set('GroupOutputVolume', 0)
```

Command	Value	Value

GroupProgramMute	'On'	'Off'	

```
# GroupProgramMute example
InterfaceName.Set('GroupProgramMute', 'On')
```

Command	Value
GroupProgramVolume	-100 to 0 in steps of 0.1

```
# GroupProgramVolume example
InterfaceName.Set('GroupProgramVolume', 0)
```

Command	Value
GroupTreble	-24 to 12 in steps of 0.1

```
# GroupTreble example
InterfaceName.Set('GroupTreble', 12)
```

Command	Value	Value
HDCPInputAuthorization	'On'	'Off'
Qualifier Key	Qualifier Value	Qualifier Value
'Input'	'3' – '6' [4]	'3' – '8' [5]

```
# HDCPInputAuthorization example
InterfaceName.Set('HDCPInputAuthorization', 'On', {'Input': '3'})
```

Command	Value	Value	
Input [9]	'1' – '6' [4]	'1' – '8' [5]	
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Type'	'Audio'	'Video'	'Audio/Video'

```
# Input example
InterfaceName.Set('Input', '0', {'Type': 'Audio'})
```

Command	Value
InputPresetRecall	'1' – '128'

```
# InputPresetRecall example
InterfaceName.Set('InputPresetRecall', '1')
```

Command	Value
InputPresetSave	'1' – '128'

```
# InputPresetSave example
InterfaceName.Set('InputPresetSave', '1')
```

Command	Value	Value	Value
InputSignalType [7]	'RGB'	'YUV'	'RGBcvS'
	'S-Video'	'Composite'	
Qualifier Key	Qualifier Value		
'Input'	'1' – '2'		

```
# InputSignalType example
InterfaceName.Set('InputSignalType', 'RGB', {'Input': '1'})
```

Command	Value	
LineInputGain	-18 to 24 in steps of 0.1	
Qualifier Key	Qualifier Value	Qualifier Value
'Input'	'1' – '6' [4]	'1' – '8' [5]
Qualifier Key	Qualifier Value	Qualifier Value
'Type'	'Analog'	'LPCM-2Ch'

```
# LineInputGain example
InterfaceName.Set('LineInputGain', 24, {'Input': '1', 'Type': 'Analog'})
```

Command	Value
MicLineInputGain	-18 to 60 in steps of 0.1
Qualifier Key	Qualifier Value
'Input'	'1' – '2'

```
# MicLineInputGain example
InterfaceName.Set('MicLineInputGain', 60, {'Input': '1'})
```

Command	Value	Value
MicLineInputMute	'On'	'Off'

Qualifier Key **'Input'**	Qualifier Value '1' – '2'		
# MicLineInputMute example `InterfaceName.Set('MicLineInputMute', 'On', {'Input': '1'})`			
Command **MicMixLevel**	Value -100 to 0 in steps of 0.1		
Qualifier Key **'Mic'**	Qualifier Value '1' – '2'		
# MicMixLevel example `InterfaceName.Set('MicMixLevel', 0, {'Mic': '1'})`			
Command **MicMixMute**	Value 'On'	Value 'Off'	
Qualifier Key **'Mic'**	Qualifier Value '1' – '2'		
# MicMixMute example `InterfaceName.Set('MicMixMute', 'On', {'Mic': '1'})`			
Command **OutputFormat**	Value 'Auto' 'HDMI RGB Limited' 'HDMI YUV 422 Full'	Value 'DVI RGB 444' 'HDMI YUV 444 Full' 'HDMI YUV 422 Limited'	Value 'HDMI RGB Full' 'HDMI YUV 444 Limited'
Qualifier Key **'Output'**	Qualifier Value 'A'	Qualifier Value 'B'	Qualifier Value 'C' [5]
# OutputFormat example `InterfaceName.Set('OutputFormat', 'Auto', {'Output': 'A'})`			
Command **OutputGain**	Value -100 to 0 in steps of 0.1		
Qualifier Key **'Output'**	Qualifier Value 'Output 1' 'Right Variable Output' 'Amplified Output' [2]	Qualifier Value 'Output 2' 'Left Digital Output' 'Left Amplified Output' [3]	Qualifier Value 'Left Variable Output' 'Right Digital Output' 'Right Amplified Output' [3]
# OutputGain example `InterfaceName.Set('OutputGain', 0, {'Output': 'Output 1'})`			
Command **OutputResolution**	Value '640x480 (50Hz)' '1024x768 (50Hz)' '1280x768 (50Hz)' '1360x765 (50Hz)' '1366x768 (50Hz)' '1400x1050 (50Hz)' '1600x1200 (50Hz)' '720p (50Hz)' '2048x1080 (50Hz)' '852x480 (60Hz)' '1024x1024 (60Hz)' '1280x1024 (60Hz)' '1365x768 (60Hz)' '1440x900 (60Hz)' '1680x1050 (60Hz)' '480p (60Hz)' '1080p (60Hz)' '800x600 (75Hz)'	Value '800x600 (50Hz)' '1024x852 (50Hz)' '1280x800 (50Hz)' '1360x768 (50Hz)' '1365x1024 (50Hz)' '1600x900 (50Hz)' '1920x1200 (50Hz)' '1080i (50Hz)' '640x480 (60Hz)' '1024x768 (60Hz)' '1280x768 (60Hz)' '1360x765 (60Hz)' '1366x768 (60Hz)' '1400x1050 (60Hz)' '1600x1200 (60Hz)' '720p (60Hz)' '2048x1080 (60Hz)' '852x480 (75Hz)'	Value '852x480 (50Hz)' '1024x1024 (50Hz)' '1280x1024 (50Hz)' '1365x768 (50Hz)' '1440x900 (50Hz)' '1680x1050 (50Hz)' '576p (50Hz)' '1080p (50Hz)' '800x600 (60Hz)' '1024x852 (60Hz)' '1280x800 (60Hz)' '1360x768 (60Hz)' '1365x1024 (60Hz)' '1600x900 (60Hz)' '1920x1200 (60Hz)' '1080i (60Hz)' '640x480 (75Hz)' '1024x768 (75Hz)'

	'1024x852 (75Hz)' '1280x800 (75Hz)' '1360x768 (75Hz)' '1365x1024 (75Hz)' '2048x1080 (23.98Hz)' '720p (25Hz)' '720p (29.97Hz)' '720p (30Hz)' '720p (59.94Hz)' '2048x1080 (59.94Hz)'	'1024x1024 (75Hz)' '1280x1024 (75Hz)' '1365x768 (75Hz)' '1440x900 (75Hz)' '1080p (24Hz)' '1080p (25Hz)' '1080p (29.97Hz)' '1080p (30Hz)' '1080i (59.94Hz)' '480p (59.94Hz)'	'1280x768 (75Hz)' '1360x765 (75Hz)' '1366x768 (75Hz)' '1080p (23.98Hz)' '2048x1080 (24Hz)' '2048x1080 (25Hz)' '2048x1080 (29.97Hz)' '2048x1080 (30Hz)' '1080p (59.94Hz)'
`# OutputResolution example` `InterfaceName.Set('OutputResolution', '640x480 (50Hz)')`			
Command **PowerSaveMode**	Value 'On'	Value 'Off'	
`# PowerSaveMode example` `InterfaceName.Set('PowerSaveMode', 'On')`			
Command **SwitchTransition**	Value 'Cut'	Value 'Fade Through Black'	
`# SwitchTransition example` `InterfaceName.Set('SwitchTransition', 'Cut')`			
Command **TestPattern**	Value 'Crop' 'Grayscale' 'Off'	Value 'Alternating Pixels' 'Blue Mode'	Value 'Color Bars' 'Audio Test'
`# TestPattern example` `InterfaceName.Set('TestPattern', 'Crop')`			
Command **UserPresetRecall**	Value '1' – '16'		
`# UserPresetRecall example` `InterfaceName.Set('UserPresetRecall', '1')`			
Command **UserPresetSave**	Value '1' – '16'		
`# UserPresetSave example` `InterfaceName.Set('UserPresetSave', '1')`			
Command **VariableAnalogOutputMicLine**	Value 'Include'	Value 'Exclude'	
`# VariableAnalogOutputMicLine example` `InterfaceName.Set('VariableAnalogOutputMicLine', 'Include')`			
Command **VariableAnalogOutputProgram**	Value 'Stereo Program'	Value 'Dual Mono Program'	Value 'No Program'
`# VariableAnalogOutputProgram example` `InterfaceName.Set('VariableAnalogOutputProgram', 'Stereo Program')`			
Command **VideoMute**	Value 'On'	Value 'Off'	Value 'On with Sync'
Qualifier Key **'Output'**	Qualifier Value 'A'	Qualifier Value 'B'	Qualifier Value 'C' [5]
`# VideoMute example` `InterfaceName.Set('VideoMute', 'On', {'Output': 'A'})`			

[1] Only supported for MA and SA models

[2] Only supported for MA models

[3] Only supported for SA models

[4] Only supported for IN1606 models
[5] Only supported for IN1608 models
[6] Inputs 1 and 2 only support "Analog" and "None" states
[7] Input Signal Type can only be selected for analog inputs 1 and 2
[8] Mode 1: Lock entire front panel Mode 2: Limit front panel control to input switching and volume control only
[9] Command will not work unless Auto Switch Mode is set to Off
[10] Reference 'Front Panel Volume Knob Control' section above

Status Available

For all commands except for HDCPInputStatus, HDCPOutputStatus, InputSignalStatus, ScreenSaverStatus, and Temperature, Update should be called only once since the command's status will be updated automatically as the device's status changes. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AspectRatio	'Fill'	'Follow'	
Qualifier Key **'Input'**	Qualifier Value '1' – '6' [4]	Qualifier Value '1' – '8' [5]	

```
# AspectRatio example
InterfaceName.Update('AspectRatio', {'Input': '1'})
Value = InterfaceName.ReadStatus('AspectRatio', {'Input': '1'})
InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)
```

Command	Value	Value	Value
AudioFormat	'Analog' 'LPCM-2Ch Auto'	'LPCM-2Ch' 'Multi-Ch Auto'	'Multi-Ch' 'None'
Qualifier Key **'Input'**	Qualifier Value '1' – '6' [4]	Qualifier Value '1' – '8' [5]	

```
# AudioFormat example
InterfaceName.Update('AudioFormat', {'Input': '1'})
Value = InterfaceName.ReadStatus('AudioFormat', {'Input': '1'})
InterfaceName.SubscribeStatus('AudioFormat', None, FeedbackHandler)
```

Command	Value	Value	Value
AudioMute	'On'	'Off'	
Qualifier Key **'Output'**	Qualifier Value 'Output 1' 'Right Variable Output' 'Left Amplified Output' [3]	Qualifier Value 'Output 2' 'Left Digital Output' 'Right Amplified Output' [3]	Qualifier Value 'Left Variable Output' 'Right Digital Output' 'Amplified Output' [2]

```
# AudioMute example
InterfaceName.Update('AudioMute', {'Output': 'Output 1'})
Value = InterfaceName.ReadStatus('AudioMute', {'Output': 'Output 1'})
InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)
```

Command	Value	Value	Value
AutoSwitchMode	'Off'	'Highest Active Input'	'Lowest Active Input'

```
# AutoSwitchMode example
InterfaceName.Update('AutoSwitchMode')
Value = InterfaceName.ReadStatus('AutoSwitchMode')
InterfaceName.SubscribeStatus('AutoSwitchMode', None, FeedbackHandler)
```

Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'

```
# ConnectionStatus example
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value	Value	Value
Ducking	'Enable'	'Disable'	
Qualifier Key: **'Mic/Line'**	Qualifier Value: '1' – '2'		

```
# Ducking example
InterfaceName.Update('Ducking', {'Mic/Line': '1'})
Value = InterfaceName.ReadStatus('Ducking', {'Mic/Line': '1'})
InterfaceName.SubscribeStatus('Ducking', None, FeedbackHandler)
```

Command	Value	Value	Value
ExecutiveMode [8]	'Off'	'Mode 1'	'Mode 2'

```
# ExecutiveMode example
InterfaceName.Update('ExecutiveMode')
Value = InterfaceName.ReadStatus('ExecutiveMode')
InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)
```

Command	Value	Value
Freeze	'On'	'Off'

```
# Freeze example
InterfaceName.Update('Freeze')
Value = InterfaceName.ReadStatus('Freeze')
InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)
```

Command	Value	Value	Value
GlobalVideoMute	'On'	'On with Sync'	'Off'

```
# GlobalVideoMute example
InterfaceName.Update('GlobalVideoMute')
Value = InterfaceName.ReadStatus('GlobalVideoMute')
InterfaceName.SubscribeStatus('GlobalVideoMute', None, FeedbackHandler)
```

Command	Value
GroupBass	-24 to 12 in steps of 0.1

```
# GroupBass example
InterfaceName.Update('GroupBass')
Value = InterfaceName.ReadStatus('GroupBass')
InterfaceName.SubscribeStatus('GroupBass', None, FeedbackHandler)
```

Command	Value	Value
GroupMicMute	'On'	'Off'

```
# GroupMicMute example
InterfaceName.Update('GroupMicMute')
Value = InterfaceName.ReadStatus('GroupMicMute')
InterfaceName.SubscribeStatus('GroupMicMute', None, FeedbackHandler)
```

Command	Value
GroupMicVolume	-100 to 0 in steps of 0.1

```
# GroupMicVolume example
InterfaceName.Update('GroupMicVolume')
Value = InterfaceName.ReadStatus('GroupMicVolume')
InterfaceName.SubscribeStatus('GroupMicVolume', None, FeedbackHandler)
```

Command	Value	Value
GroupOutputMute	'On'	'Off'

```
# GroupOutputMute example
InterfaceName.Update('GroupOutputMute')
Value = InterfaceName.ReadStatus('GroupOutputMute')
InterfaceName.SubscribeStatus('GroupOutputMute', None, FeedbackHandler)
```

Command	Value
GroupOutputVolume [1]	-100 to 0 in steps of 0.1

```
# GroupOutputVolume example
InterfaceName.Update('GroupOutputVolume')
```

```
Value = InterfaceName.ReadStatus('GroupOutputVolume')
InterfaceName.SubscribeStatus('GroupOutputVolume', None, FeedbackHandler)
```

Command	Value	Value	
GroupProgramMute	'On'	'Off'	

```
# GroupProgramMute example
InterfaceName.Update('GroupProgramMute')
Value = InterfaceName.ReadStatus('GroupProgramMute')
InterfaceName.SubscribeStatus('GroupProgramMute', None, FeedbackHandler)
```

Command	Value
GroupProgramVolume	-100 to 0 in steps of 0.1

```
# GroupProgramVolume example
InterfaceName.Update('GroupProgramVolume')
Value = InterfaceName.ReadStatus('GroupProgramVolume')
InterfaceName.SubscribeStatus('GroupProgramVolume', None, FeedbackHandler)
```

Command	Value
GroupTreble	-24 to 12 in steps of 0.1

```
# GroupTreble example
InterfaceName.Update('GroupTreble')
Value = InterfaceName.ReadStatus('GroupTreble')
InterfaceName.SubscribeStatus('GroupTreble', None, FeedbackHandler)
```

Command	Value	Value	
HDCPInputAuthorization	'On'	'Off'	
Qualifier Key	Qualifier Value	Qualifier Value	
'Input'	'3' – '6' [4]	'3' – '8' [5]	

```
# HDCPInputAuthorization example
InterfaceName.Update('HDCPInputAuthorization', {'Input': '3'})
Value = InterfaceName.ReadStatus('HDCPInputAuthorization', {'Input': '3'})
InterfaceName.SubscribeStatus('HDCPInputAuthorization', None, FeedbackHandler)
```

Command	Value	Value	Value
HDCPInputStatus	'No Source Device Detected'	'Source Detected with HDCP'	'Source Detected without HDCP'
Qualifier Key	Qualifier Value	Qualifier Value	
'Input'	'3' – '6' [4]	'3' – '8' [5]	

```
# HDCPInputStatus example
InterfaceName.Update('HDCPInputStatus', {'Input': '3'})
Value = InterfaceName.ReadStatus('HDCPInputStatus', {'Input': '3'})
InterfaceName.SubscribeStatus('HDCPInputStatus', None, FeedbackHandler)
```

Command	Value	Value	Value
HDCPOutputStatus	'No Sink Device Detected'	'Sink Detected with HDCP'	'Sink Detected without HDCP'
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Output'	'A'	'B'	'C' [5]

```
# HDCPOutputStatus example
InterfaceName.Update('HDCPOutputStatus', {'Output': 'A'})
Value = InterfaceName.ReadStatus('HDCPOutputStatus', {'Output': 'A'})
InterfaceName.SubscribeStatus('HDCPOutputStatus', None, FeedbackHandler)
```

Command	Value	Value	
Input	'0' – '6' [4]	'0' – '8' [5]	
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Type'	'Audio'	'Video'	'Audio/Video'

```
# Input example
InterfaceName.Update('Input', {'Type': 'Audio'})
Value = InterfaceName.ReadStatus('Input', {'Type': 'Audio'})
InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)
```

Command	Value	Value
InputSignalStatus	'Active'	'Not Active'

Qualifier Key **'Input'**	Qualifier Value '1' – '6' [4]	Qualifier Value '1' – '8' [5]	

```
# InputSignalStatus example
InterfaceName.Update('InputSignalStatus', {'Input': '1'})
Value = InterfaceName.ReadStatus('InputSignalStatus', {'Input': '1'})
InterfaceName.SubscribeStatus('InputSignalStatus', None, FeedbackHandler)
```

Command **InputSignalType**	Value 'RGB' 'S-Video'	Value 'YUV' 'Composite'	Value 'RGBcvS'
Qualifier Key **'Input'**	Qualifier Value '1' – '2'		

```
# InputSignalType example
InterfaceName.Update('InputSignalType', {'Input': '1'})
Value = InterfaceName.ReadStatus('InputSignalType', {'Input': '1'})
InterfaceName.SubscribeStatus('InputSignalType', None, FeedbackHandler)
```

Command **LineInputGain**	Value -18 to 24 in steps of 0.1		
Qualifier Key **'Input'**	Qualifier Value '1' – '6' [4]	Qualifier Value '1' – '8' [5]	
Qualifier Key **'Type'**	Qualifier Value 'Analog'	Qualifier Value 'LPCM-2Ch'	

```
# LineInputGain example
InterfaceName.Update('LineInputGain', {'Input': '1', 'Type': 'Analog'})
Value = InterfaceName.ReadStatus('LineInputGain', {'Input': '1', 'Type': 'Analog'})
InterfaceName.SubscribeStatus('LineInputGain', None, FeedbackHandler)
```

Command **MicLineInputGain**	Value -18 to 60 in steps of 0.1		
Qualifier Key **'Input'**	Qualifier Value '1' – '2'		

```
# MicLineInputGain example
InterfaceName.Update('MicLineInputGain', {'Input': '1'})
Value = InterfaceName.ReadStatus('MicLineInputGain', {'Input': '1'})
InterfaceName.SubscribeStatus('MicLineInputGain', None, FeedbackHandler)
```

Command **MicLineInputMute**	Value 'On'	Value 'Off'	
Qualifier Key **'Input'**	Qualifier Value '1' – '2'		

```
# MicLineInputMute example
InterfaceName.Update('MicLineInputMute', {'Input': '1'})
Value = InterfaceName.ReadStatus('MicLineInputMute', {'Input': '1'})
InterfaceName.SubscribeStatus('MicLineInputMute', None, FeedbackHandler)
```

Command **MicMixLevel**	Value -100 to 0 in steps of 0.1		
Qualifier Key **'Mic'**	Qualifier Value '1' – '2'		

```
# MicMixLevel example
InterfaceName.Update('MicMixLevel', {'Mic': '1'})
Value = InterfaceName.ReadStatus('MicMixLevel', {'Mic': '1'})
InterfaceName.SubscribeStatus('MicMixLevel', None, FeedbackHandler)
```

Command **MicMixMute**	Value 'On'	Value 'Off'	
Qualifier Key **'Mic'**	Qualifier Value '1' – '2'		

```
# MicMixMute example
InterfaceName.Update('MicMixMute', {'Mic': '1'})
Value = InterfaceName.ReadStatus('MicMixMute', {'Mic': '1'})
```

```
InterfaceName.SubscribeStatus('MicMixMute', None, FeedbackHandler)
```

Command	Value	Value	Value
OutputFormat	'Auto'	'DVI RGB 444'	'HDMI RGB Full'
	'HDMI RGB Limited'	'HDMI YUV 444 Full'	'HDMI YUV 444 Limited'
	'HDMI YUV 422 Full'	'HDMI YUV 422 Limited'	
Qualifier Key	**Qualifier Value**	**Qualifier Value**	**Qualifier Value**
'Output'	'A'	'B'	'C'

```
# OutputFormat example
InterfaceName.Update('OutputFormat', {'Output': 'A'})
Value = InterfaceName.ReadStatus('OutputFormat', {'Output': 'A'})
InterfaceName.SubscribeStatus('OutputFormat', None, FeedbackHandler)
```

Command	Value		
OutputGain	-100 to 0 in steps of 0.1		
Qualifier Key	**Qualifier Value**	**Qualifier Value**	**Qualifier Value**
'Output'	'Output 1'	'Output 2'	'Left Variable Output'
	'Right Variable Output'	'Left Digital Output'	'Right Digital Output'
	'Amplified Output'	'Left Amplified Output'	'Right Amplified Output'

```
# OutputGain example
InterfaceName.Update('OutputGain', {'Output': 'Output 1'})
Value = InterfaceName.ReadStatus('OutputGain', {'Output': 'Output 1'})
InterfaceName.SubscribeStatus('OutputGain', None, FeedbackHandler)
```

Command	Value	Value	Value
OutputResolution	'640x480 (50Hz)'	'800x600 (50Hz)'	'852x480 (50Hz)'
	'1024x768 (50Hz)'	'1024x852 (50Hz)'	'1024x1024 (50Hz)'
	'1280x768 (50Hz)'	'1280x800 (50Hz)'	'1280x1024 (50Hz)'
	'1360x765 (50Hz)'	'1360x768 (50Hz)'	'1365x768 (50Hz)'
	'1366x768 (50Hz)'	'1365x1024 (50Hz)'	'1440x900 (50Hz)'
	'1400x1050 (50Hz)'	'1600x900 (50Hz)'	'1680x1050 (50Hz)'
	'1600x1200 (50Hz)'	'1920x1200 (50Hz)'	'576p (50Hz)'
	'720p (50Hz)'	'1080i (50Hz)'	'1080p (50Hz)'
	'2048x1080 (50Hz)'	'640x480 (60Hz)'	'800x600 (60Hz)'
	'852x480 (60Hz)'	'1024x768 (60Hz)'	'1024x852 (60Hz)'
	'1024x1024 (60Hz)'	'1280x768 (60Hz)'	'1280x800 (60Hz)'
	'1280x1024 (60Hz)'	'1360x765 (60Hz)'	'1360x768 (60Hz)'
	'1365x768 (60Hz)'	'1366x768 (60Hz)'	'1365x1024 (60Hz)'
	'1440x900 (60Hz)'	'1400x1050 (60Hz)'	'1600x900 (60Hz)'
	'1680x1050 (60Hz)'	'1600x1200 (60Hz)'	'1920x1200 (60Hz)'
	'480p (60Hz)'	'720p (60Hz)'	'1080i (60Hz)'
	'1080p (60Hz)'	'2048x1080 (60Hz)'	'640x480 (75Hz)'
	'800x600 (75Hz)'	'852x480 (75Hz)'	'1024x768 (75Hz)'
	'1024x852 (75Hz)'	'1024x1024 (75Hz)'	'1280x768 (75Hz)'
	'1280x800 (75Hz)'	'1280x1024 (75Hz)'	'1360x765 (75Hz)'
	'1360x768 (75Hz)'	'1365x768 (75Hz)'	'1366x768 (75Hz)'
	'1365x1024 (75Hz)'	'1440x900 (75Hz)'	'1080p (23.98Hz)'
	'2048x1080 (23.98Hz)'	'1080p (24Hz)'	'2048x1080 (24Hz)'
	'720p (25Hz)'	'1080p (25Hz)'	'2048x1080 (25Hz)'
	'720p (29.97Hz)'	'1080p (29.97Hz)'	'2048x1080 (29.97Hz)'
	'720p (30Hz)'	'1080p (30Hz)'	'2048x1080 (30Hz)'
	'720p (59.94Hz)'	'1080i (59.94Hz)'	'1080p (59.94Hz)'
	'2048x1080 (59.94Hz)'	'480p (59.94Hz)'	

```
# OutputResolution example
InterfaceName.Update('OutputResolution')
```

```
    Value = InterfaceName.ReadStatus('OutputResolution')
    InterfaceName.SubscribeStatus('OutputResolution', None, FeedbackHandler)
```

Command	Value	Value	
PowerSaveMode	'On'	'Off'	

```
    # PowerSaveMode example
    InterfaceName.Update('PowerSaveMode')
    Value = InterfaceName.ReadStatus('PowerSaveMode')
    InterfaceName.SubscribeStatus('PowerSaveMode', None, FeedbackHandler)
```

Command	Value	Value	Value
ScreenSaverStatus	'Active Input Detected; Timer not running'	'No Active Input; Timer expired; Output sync disabled'	'No Active Input; Timer running; Output sync enabled'

```
    # ScreenSaverStatus example
    InterfaceName.Update('ScreenSaverStatus')
    Value = InterfaceName.ReadStatus('ScreenSaverStatus')
    InterfaceName.SubscribeStatus('ScreenSaverStatus', None, FeedbackHandler)
```

Command	Value	Value	
SwitchTransition	'Cut'	'Fade Through Black'	

```
    # SwitchTransition example
    InterfaceName.Update('SwitchTransition')
    Value = InterfaceName.ReadStatus('SwitchTransition')
    InterfaceName.SubscribeStatus('SwitchTransition', None, FeedbackHandler)
```

Command	Value
Temperature	Degrees Celsius

```
    # Temperature example
    InterfaceName.Update('Temperature')
    Value = InterfaceName.ReadStatus('Temperature')
    InterfaceName.SubscribeStatus('Temperature', None, FeedbackHandler)
```

Command	Value	Value	Value
TestPattern	'Crop' 'Grayscale' 'Off'	'Alternating Pixels' 'Blue Mode'	'Color Bars' 'Audio Test'

```
    # TestPattern example
    InterfaceName.Update('TestPattern')
    Value = InterfaceName.ReadStatus('TestPattern')
    InterfaceName.SubscribeStatus('TestPattern', None, FeedbackHandler)
```

Command	Value	Value	Value
VideoMute	'On'	'Off'	'On with Sync'
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Output'	'A'	'B'	'C' [5]

```
    # VideoMute example
    InterfaceName.Update('VideoMute', {'Output': 'A'})
    Value = InterfaceName.ReadStatus('VideoMute', {'Output': 'A'})
    InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)
```

[1] Reference 'Front Panel Volume Knob Control' section above
[2] Only supported for MA models
[3] Only supported for SA models
[4] Only supported for IN1606 models
[5] Only supported for IN1608 models
[6] Inputs 1 and 2 only support "Analog" and "None" states
[7] Input Signal Type can only be selected for analog inputs 1 and 2
[8] Mode 1: Lock entire front panel Mode 2: Limit front panel control to input switching and volume control only
[9] Command will not work unless Auto Switch Mode is set to Off
[10] When the audio input is different than the video input, Audio/Video Type status will show “0”

Cable and Adapter Requirements

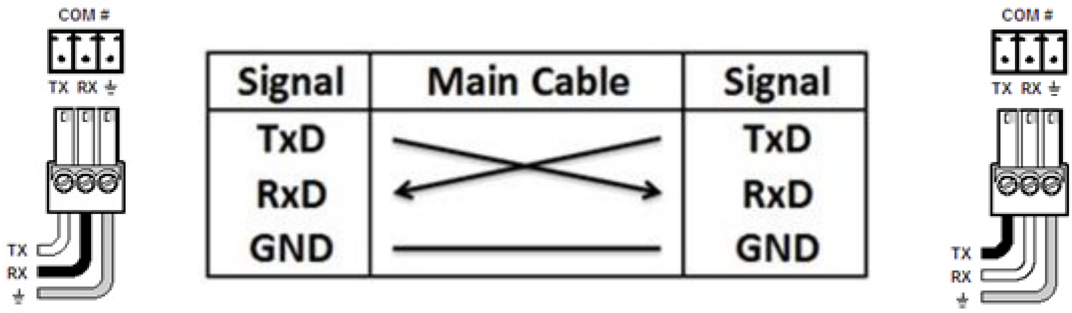
Captive Screw to Captive Screw

Notes for the Device

Serial communication

Port Type:	RS-232	**Parity:**	None
Baud Rate:	9600	**Stop Bits:**	One
Data Bits:	8	**Flow Control:**	None

Pin Assignments Diagram



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

Port Type:	Ethernet
Default Port:	22023
Logon Credentials Supported:	Yes
Default Username:	admin
Default Password:	
Multi-Connection Capabilities:	Yes
Port Changeability:	Yes

Ethernet Module Configuration Description

- Please refer to user manual for settings and changes to the network communication
- Default Username is “admin” but can be changed to “user”
    - If User password is used for authentication, control of the device may be limited.

Notes for the Device

Appendix A. Set Commands

Amplified Output Mic Line Exclude	\x1Bm20808*1au\x0D\x1Bm20809*1au\x0D\x1Bm20908*1au\x0D\x1Bm20909*1au\x0D
Amplified Output Mic Line Include	\x1Bm20808*0au\x0D\x1Bm20809*0au\x0D\x1Bm20908*0au\x0D\x1Bm20909*0au\x0D
Amplified Output Program Dual Mono Program	\x1Bn50008*1au\x0D\x1Bn50009*1au\x0D\x1Bm50008*0au\x0D\x1Bm50009*0au\x0D
Amplified Output Program No Program	\x1Bm50008*1au\x0D\x1Bm50009*1au\x0D
Amplified Output Program Stereo Program	\x1Bn50008*0au\x0D\x1Bn50009*0au\x0D\x1Bm50008*0au\x0D\x1Bm50009*0au\x0D
Analog Output 1 Program L+R Program (Mono)	\x1Bn50000*1au\x0D\x1Bm50000*0au\x0D\x1Bm50001*0au\x0D
Analog Output 1 Program Left Program	\x1Bn50000*0au\x0D\x1Bm50000*0au\x0D\x1Bm50001*1au\x0D
Analog Output 1 Program No Program	\x1Bm50000*1au\x0D\x1Bm50001*1au\x0D
Analog Output 2 Program L+R Program (Mono)	\x1Bn50002*1au\x0D\x1Bm50002*0au\x0D\x1Bm50003*0au\x0D
Analog Output 2 Program No Program	\x1Bm50002*1au\x0D\x1Bm50003*1au\x0D
Analog Output 2 Program Right Program	\x1Bn50002*0au\x0D\x1Bm50002*1au\x0D\x1Bm50003*0au\x0D
Analog Output Mic Line Exclude Output 1	\x1Bm20800*1au\x0D\x1Bm20900*1au\x0D
Analog Output Mic Line Exclude Output 2	\x1Bm20802*1au\x0D\x1Bm20902*1au\x0D
Analog Output Mic Line Include Output 1	\x1Bm20800*0au\x0D\x1Bm20900*0au\x0D
Analog Output Mic Line Include Output 2	\x1Bm20802*0au\x0D\x1Bm20902*0au\x0D
Aspect Ratio Fill Input 1	w1*1ASPR\x0D
Aspect Ratio Fill Input 6	w6*1ASPR\x0D
Aspect Ratio Fill Input 8	w8*1ASPR\x0D
Aspect Ratio Follow Input 1	w1*2ASPR\x0D
Aspect Ratio Follow Input 6	w6*2ASPR\x0D
Aspect Ratio Follow Input 8	w8*2ASPR\x0D
Audio Format Analog Input 1	wI1*1AFMT\x0D
Audio Format Analog Input 6	wI6*1AFMT\x0D
Audio Format Analog Input 8	wI8*1AFMT\x0D
Audio Format LPCM-2Ch Auto Input 6	wI6*4AFMT\x0D
Audio Format LPCM-2Ch Auto Input 8	wI8*4AFMT\x0D
Audio Format LPCM-2Ch Input 6	wI6*2AFMT\x0D
Audio Format LPCM-2Ch Input 8	wI8*2AFMT\x0D
Audio Format Multi-Ch Auto Input 6	wI6*5AFMT\x0D
Audio Format Multi-Ch Auto Input 8	wI8*5AFMT\x0D
Audio Format Multi-Ch Input 6	wI6*3AFMT\x0D
Audio Format Multi-Ch Input 8	wI8*3AFMT\x0D
Audio Format None Input 1	wI1*0AFMT\x0D
Audio Format None Input 6	wI6*0AFMT\x0D
Audio Format None Input 8	wI8*0AFMT\x0D
Audio Mute Off Output Amplified Output	wM60008*0AU\x0D
Audio Mute Off Output Left Amplified Output	wM60008*0AU\x0D
Audio Mute Off Output Left Digital Output	wM60006*0AU\x0D

Audio Mute Off Output Left Variable Output	wM60004*0AU\x0D
Audio Mute Off Output Output 1	wM60000*0AU\x0D
Audio Mute Off Output Output 2	wM60002*0AU\x0D
Audio Mute Off Output Right Amplified Output	wM60009*0AU\x0D
Audio Mute Off Output Right Digital Output	wM60007*0AU\x0D
Audio Mute Off Output Right Variable Output	wM60005*0AU\x0D
Audio Mute On Output Amplified Output	wM60008*1AU\x0D
Audio Mute On Output Left Amplified Output	wM60008*1AU\x0D
Audio Mute On Output Left Digital Output	wM60006*1AU\x0D
Audio Mute On Output Left Variable Output	wM60004*1AU\x0D
Audio Mute On Output Output 1	wM60000*1AU\x0D
Audio Mute On Output Output 2	wM60002*1AU\x0D
Audio Mute On Output Right Amplified Output	wM60009*1AU\x0D
Audio Mute On Output Right Digital Output	wM60007*1AU\x0D
Audio Mute On Output Right Variable Output	wM60005*1AU\x0D
Auto Image Execute	0*A
Auto Image Execute and Fill	1*A
Auto Image Execute and Follow	2*A
Auto Switch Mode Highest Active Input	w1AUSW\x0D
Auto Switch Mode Lowest Active Input	w2AUSW\x0D
Auto Switch Mode Off	w0AUSW\x0D
Digital Output Mic Line Exclude	\x1Bm20806*1au\x0D\x1Bm20807*1au\x0D\x1Bm20906*1au\x0D\x1Bm20907*1au\x0D
Digital Output Mic Line Include	\x1Bm20806*0au\x0D\x1Bm20807*0au\x0D\x1Bm20906*0au\x0D\x1Bm20907*0au\x0D
Digital Output Program Dual Mono Program	\x1Bn50006*1au\x0D\x1Bn50007*1au\x0D\x1Bm50006*0au\x0D\x1Bm50007*0au\x0D
Digital Output Program No Program	\x1Bm50006*1au\x0D\x1Bm50007*1au\x0D
Digital Output Program Stereo Program	\x1Bn50006*0au\x0D\x1Bn50007*0au\x0D\x1Bm50006*0au\x0D\x1Bm50007*0au\x0D
Ducking Disable Mic/Line 1	we44800*0au\x0D
Ducking Disable Mic/Line 2	we44801*0au\x0D
Ducking Enable Mic/Line 1	we44800*1au\x0D
Ducking Enable Mic/Line 2	we44801*1au\x0D
Executive Mode Mode 1	1X
Executive Mode Mode 2	2X
Executive Mode Off	0X
Freeze Off	0F
Freeze On	1F
Global Video Mute Off	0B
Global Video Mute On	1B
Global Video Mute On with Sync	2B
Group Bass 12	wD5*+120GRPM\x0D
Group Bass -24	wD5*-240GRPM\x0D
Group Mic Mute Off	wD4*0GRPM\x0D
Group Mic Mute On	wD4*1GRPM\x0D

Group Mic Volume 0	wD3*+00000GRPM\x0D
Group Mic Volume -100	wD3*-01000GRPM\x0D
Group Output Mute Off	wD7*0GRPM\x0D
Group Output Mute On	wD7*1GRPM\x0D
Group Output Volume 0	wD8*+00000GRPM\x0D
Group Output Volume -100	wD8*-01000GRPM\x0D
Group Program Mute Off	wD2*0GRPM\x0D
Group Program Mute On	wD2*1GRPM\x0D
Group Program Volume 0	wD1*+00000GRPM\x0D
Group Program Volume -100	wD1*-01000GRPM\x0D
Group Treble 12	wD6*+120GRPM\x0D
Group Treble -24	wD6*-240GRPM\x0D
HDCP Input Authorization Off Input 3	wE3*0HDCP\x0D
HDCP Input Authorization Off Input 6	wE6*0HDCP\x0D
HDCP Input Authorization Off Input 8	wE8*0HDCP\x0D
HDCP Input Authorization On Input 3	wE3*1HDCP\x0D
HDCP Input Authorization On Input 6	wE6*1HDCP\x0D
HDCP Input Authorization On Input 8	wE8*1HDCP\x0D
Input 1 Type Audio	1$
Input 1 Type Audio/Video	1!
Input 1 Type Video	1&
Input 6 Type Audio	6$
Input 6 Type Audio/Video	6!
Input 6 Type Video	6&
Input 8 Type Audio	8$
Input 8 Type Audio/Video	8!
Input 8 Type Video	8&
Input Preset Recall 1	2*1.
Input Preset Recall 128	2*128.
Input Preset Save 1	2*1,
Input Preset Save 128	2*128,
Input Signal Type Composite Input 1	1*5\
Input Signal Type Composite Input 2	2*5\
Input Signal Type RGB Input 1	1*1\
Input Signal Type RGB Input 2	2*1\
Input Signal Type RGBcvS Input 1	1*3\
Input Signal Type RGBcvS Input 2	2*3\
Input Signal Type S-Video Input 1	1*4\
Input Signal Type S-Video Input 2	2*4\
Input Signal Type YUV Input 1	1*2\
Input Signal Type YUV Input 2	2*2\
Line Input Gain -18 Input 1 Type Analog	wG30000*-180AU\x0D
Line Input Gain -18 Input 1 Type LPCM-2Ch	wD30100*-180AU\x0D
Line Input Gain -18 Input 6 Type Analog	wG30010*-180AU\x0D

Line Input Gain -18 Input 6 Type LPCM-2Ch	wD30110*-180AU\x0D
Line Input Gain -18 Input 8 Type Analog	wG30014*-180AU\x0D
Line Input Gain -18 Input 8 Type LPCM-2Ch	wD30114*-180AU\x0D
Line Input Gain 24 Input 1 Type Analog	wG30000*240AU\x0D
Line Input Gain 24 Input 1 Type LPCM-2Ch	wD30100*240AU\x0D
Line Input Gain 24 Input 6 Type Analog	wG30010*240AU\x0D
Line Input Gain 24 Input 6 Type LPCM-2Ch	wD30110*240AU\x0D
Line Input Gain 24 Input 8 Type Analog	wG30014*240AU\x0D
Line Input Gain 24 Input 8 Type LPCM-2Ch	wD30114*240AU\x0D
Mic Line Input Gain -18 Input 1	wG40000*-180AU\x0D
Mic Line Input Gain -18 Input 2	wG40001*-180AU\x0D
Mic Line Input Gain 60 Input 1	wG40000*600AU\x0D
Mic Line Input Gain 60 Input 2	wG40001*600AU\x0D
Mic Line Input Mute Off Input 1	wM40000*0AU\x0D
Mic Line Input Mute Off Input 2	wM40001*0AU\x0D
Mic Line Input Mute On Input 1	wM40000*1AU\x0D
Mic Line Input Mute On Input 2	wM40001*1AU\x0D
Mic Mix Level 0 Mic 1	wG40100*0AU\x0D
Mic Mix Level 0 Mic 2	wG40101*0AU\x0D
Mic Mix Level -100 Mic 1	wG40100*-1000AU\x0D
Mic Mix Level -100 Mic 2	wG40101*-1000AU\x0D
Mic Mix Mute Off Mic 1	wM40000*0AU\x0D
Mic Mix Mute Off Mic 2	wM40001*0AU\x0D
Mic Mix Mute On Mic 1	wM40000*1AU\x0D
Mic Mix Mute On Mic 2	wM40001*1AU\x0D
Output Format Auto Output A	w1*0VTPO\x0D
Output Format Auto Output B	w2*0VTPO\x0D
Output Format Auto Output C	w3*0VTPO\x0D
Output Format DVI RGB 444 Output A	w1*1VTPO\x0D
Output Format DVI RGB 444 Output B	w2*1VTPO\x0D
Output Format DVI RGB 444 Output C	w3*1VTPO\x0D
Output Format HDMI RGB Full Output A	w1*2VTPO\x0D
Output Format HDMI RGB Full Output B	w2*2VTPO\x0D
Output Format HDMI RGB Full Output C	w3*2VTPO\x0D
Output Format HDMI RGB Limited Output A	w1*3VTPO\x0D
Output Format HDMI RGB Limited Output B	w2*3VTPO\x0D
Output Format HDMI RGB Limited Output C	w3*3VTPO\x0D
Output Format HDMI YUV 422 Full Output A	w1*6VTPO\x0D
Output Format HDMI YUV 422 Full Output B	w2*6VTPO\x0D
Output Format HDMI YUV 422 Full Output C	w3*6VTPO\x0D
Output Format HDMI YUV 422 Limited Output A	w1*7VTPO\x0D
Output Format HDMI YUV 422 Limited Output B	w2*7VTPO\x0D
Output Format HDMI YUV 422 Limited Output C	w3*7VTPO\x0D
Output Format HDMI YUV 444 Full Output A	w1*4VTPO\x0D

Output Format HDMI YUV 444 Full Output B	w2*4VTPO\x0D
Output Format HDMI YUV 444 Full Output C	w3*4VTPO\x0D
Output Format HDMI YUV 444 Limited Output A	w1*5VTPO\x0D
Output Format HDMI YUV 444 Limited Output B	w2*5VTPO\x0D
Output Format HDMI YUV 444 Limited Output C	w3*5VTPO\x0D
Output Gain 0 Output Amplified Output	wG60008*0AU\x0D
Output Gain 0 Output Left Amplified Output	wG60008*0AU\x0D
Output Gain 0 Output Left Digital Output	wG60006*0AU\x0D
Output Gain 0 Output Left Variable Output	wG60004*0AU\x0D
Output Gain 0 Output Output 1	wG60000*0AU\x0D
Output Gain 0 Output Output 2	wG60002*0AU\x0D
Output Gain 0 Output Right Amplified Output	wG60009*0AU\x0D
Output Gain 0 Output Right Digital Output	wG60007*0AU\x0D
Output Gain 0 Output Right Variable Output	wG60005*0AU\x0D
Output Gain -100 Output Amplified Output	wG60008*-1000AU\x0D
Output Gain -100 Output Left Amplified Output	wG60008*-1000AU\x0D
Output Gain -100 Output Left Digital Output	wG60006*-1000AU\x0D
Output Gain -100 Output Left Variable Output	wG60004*-1000AU\x0D
Output Gain -100 Output Output 1	wG60000*-1000AU\x0D
Output Gain -100 Output Output 2	wG60002*-1000AU\x0D
Output Gain -100 Output Right Amplified Output	wG60009*-1000AU\x0D
Output Gain -100 Output Right Digital Output	wG60007*-1000AU\x0D
Output Gain -100 Output Right Variable Output	wG60005*-1000AU\x0D
Output Resolution 1024x1024 (50Hz)	w25RATE\x0D
Output Resolution 1024x1024 (60Hz)	w26RATE\x0D
Output Resolution 1024x1024 (75Hz)	w27RATE\x0D
Output Resolution 1024x768 (50Hz)	w19RATE\x0D
Output Resolution 1024x768 (60Hz)	w20RATE\x0D
Output Resolution 1024x768 (75Hz)	w21RATE\x0D
Output Resolution 1024x852 (50Hz)	w22RATE\x0D
Output Resolution 1024x852 (60Hz)	w23RATE\x0D
Output Resolution 1024x852 (75Hz)	w24RATE\x0D
Output Resolution 1080i (50Hz)	w74RATE\x0D
Output Resolution 1080i (59.94Hz)	w75RATE\x0D
Output Resolution 1080i (60Hz)	w76RATE\x0D
Output Resolution 1080p (23.98Hz)	w77RATE\x0D
Output Resolution 1080p (24Hz)	w78RATE\x0D
Output Resolution 1080p (25Hz)	w79RATE\x0D
Output Resolution 1080p (29.97Hz)	w80RATE\x0D
Output Resolution 1080p (30Hz)	w81RATE\x0D
Output Resolution 1080p (50Hz)	w82RATE\x0D
Output Resolution 1080p (59.94Hz)	w83RATE\x0D
Output Resolution 1080p (60Hz)	w84RATE\x0D
Output Resolution 1280x1024 (50Hz)	w34RATE\x0D

Output Resolution 1280x1024 (60Hz)	w35RATE\x0D
Output Resolution 1280x1024 (75Hz)	w36RATE\x0D
Output Resolution 1280x768 (50Hz)	w28RATE\x0D
Output Resolution 1280x768 (60Hz)	w29RATE\x0D
Output Resolution 1280x768 (75Hz)	w30RATE\x0D
Output Resolution 1280x800 (50Hz)	w31RATE\x0D
Output Resolution 1280x800 (60Hz)	w32RATE\x0D
Output Resolution 1280x800 (75Hz)	w33RATE\x0D
Output Resolution 1360x765 (50Hz)	w37RATE\x0D
Output Resolution 1360x765 (60Hz)	w38RATE\x0D
Output Resolution 1360x765 (75Hz)	w39RATE\x0D
Output Resolution 1360x768 (50Hz)	w40RATE\x0D
Output Resolution 1360x768 (60Hz)	w41RATE\x0D
Output Resolution 1360x768 (75Hz)	w42RATE\x0D
Output Resolution 1365x1024 (50Hz)	w49RATE\x0D
Output Resolution 1365x1024 (60Hz)	w50RATE\x0D
Output Resolution 1365x1024 (75Hz)	w51RATE\x0D
Output Resolution 1365x768 (50Hz)	w43RATE\x0D
Output Resolution 1365x768 (60Hz)	w44RATE\x0D
Output Resolution 1365x768 (75Hz)	w45RATE\x0D
Output Resolution 1366x768 (50Hz)	w46RATE\x0D
Output Resolution 1366x768 (60Hz)	w47RATE\x0D
Output Resolution 1366x768 (75Hz)	w48RATE\x0D
Output Resolution 1400x1050 (50Hz)	w55RATE\x0D
Output Resolution 1400x1050 (60Hz)	w56RATE\x0D
Output Resolution 1440x900 (50Hz)	w52RATE\x0D
Output Resolution 1440x900 (60Hz)	w53RATE\x0D
Output Resolution 1440x900 (75Hz)	w54RATE\x0D
Output Resolution 1600x1200 (50Hz)	w61RATE\x0D
Output Resolution 1600x1200 (60Hz)	w62RATE\x0D
Output Resolution 1600x900 (50Hz)	w57RATE\x0D
Output Resolution 1600x900 (60Hz)	w58RATE\x0D
Output Resolution 1680x1050 (50Hz)	w59RATE\x0D
Output Resolution 1680x1050 (60Hz)	w60RATE\x0D
Output Resolution 1920x1200 (50Hz)	w63RATE\x0D
Output Resolution 1920x1200 (60Hz)	w64RATE\x0D
Output Resolution 2048x1080 (23.98Hz)	w85RATE\x0D
Output Resolution 2048x1080 (24Hz)	w86RATE\x0D
Output Resolution 2048x1080 (25Hz)	w87RATE\x0D
Output Resolution 2048x1080 (29.97Hz)	w88RATE\x0D
Output Resolution 2048x1080 (30Hz)	w89RATE\x0D
Output Resolution 2048x1080 (50Hz)	w90RATE\x0D
Output Resolution 2048x1080 (59.94Hz)	w91RATE\x0D
Output Resolution 2048x1080 (60Hz)	w92RATE\x0D

Output Resolution 480p (59.94Hz)	w65RATE\x0D
Output Resolution 480p (60Hz)	w66RATE\x0D
Output Resolution 576p (50Hz)	w67RATE\x0D
Output Resolution 640x480 (50Hz)	w10RATE\x0D
Output Resolution 640x480 (60Hz)	w11RATE\x0D
Output Resolution 640x480 (75Hz)	w12RATE\x0D
Output Resolution 720p (25Hz)	w68RATE\x0D
Output Resolution 720p (29.97Hz)	w69RATE\x0D
Output Resolution 720p (30Hz)	w70RATE\x0D
Output Resolution 720p (50Hz)	w71RATE\x0D
Output Resolution 720p (59.94Hz)	w72RATE\x0D
Output Resolution 720p (60Hz)	w73RATE\x0D
Output Resolution 800x600 (50Hz)	w13RATE\x0D
Output Resolution 800x600 (60Hz)	w14RATE\x0D
Output Resolution 800x600 (75Hz)	w15RATE\x0D
Output Resolution 852x480 (50Hz)	w16RATE\x0D
Output Resolution 852x480 (60Hz)	w17RATE\x0D
Output Resolution 852x480 (75Hz)	w18RATE\x0D
Power Save Mode Off	w0PSAV\x0D
Power Save Mode On	w1PSAV\x0D
Switch Transition Cut	w0SWEF\x0D
Switch Transition Fade Through Black	w1SWEF\x0D
Test Pattern Alternating Pixels	w2TEST\x0D
Test Pattern Audio Test	w6TEST\x0D
Test Pattern Blue Mode	w5TEST\x0D
Test Pattern Color Bars	w3TEST\x0D
Test Pattern Crop	w1TEST\x0D
Test Pattern Grayscale	w4TEST\x0D
Test Pattern Off	w0TEST\x0D
User Preset Recall 1	1*1.
User Preset Recall 16	1*16.
User Preset Save 1	1*1,
User Preset Save 16	1*16,
Variable Analog Output Mic Line Exclude	\x1Bm20804*1au\x0D\x1Bm20805*1au\x0D\x1Bm20904*1au\x0D\x1Bm20905*1au\x0D
Variable Analog Output Mic Line Include	\x1Bm20804*0au\x0D\x1Bm20805*0au\x0D\x1Bm20904*0au\x0D\x1Bm20905*0au\x0D
Variable Analog Output Program Dual Mono Program	\x1Bn50004*1au\x0D\x1Bn50005*1au\x0D\x1Bm50004*0au\x0D\x1Bm50005*0au\x0D
Variable Analog Output Program No Program	\x1Bm50004*1au\x0D\x1Bm50005*1au\x0D
Variable Analog Output Program Stereo Program	\x1Bn50004*0au\x0D\x1Bn50005*0au\x0D\x1Bm50004*0au\x0D\x1Bm50005*0au\x0D
Video Mute Off Output A	1*0B
Video Mute Off Output B	2*0B
Video Mute Off Output C	3*0B
Video Mute On Output A	1*1B

Video Mute On Output B	2*1B
Video Mute On Output C	3*1B
Video Mute On with Sync Output A	1*2B
Video Mute On with Sync Output B	2*2B
Video Mute On with Sync Output C	3*2B

Appendix B. Update Commands

Aspect Ratio Input 1	w1ASPR\x0D
Aspect Ratio Input 6	w6ASPR\x0D
Aspect Ratio Input 8	w8ASPR\x0D
Audio Format Input 1	wI1AFMT\x0D
Audio Format Input 6	wI6AFMT\x0D
Audio Format Input 8	wI8AFMT\x0D
Audio Mute Output Amplified Output	wM60008AU\x0D
Audio Mute Output Left Amplified Output	wM60008AU\x0D
Audio Mute Output Left Digital Output	wM60006AU\x0D
Audio Mute Output Left Variable Output	wM60004AU\x0D
Audio Mute Output Output 1	wM60000AU\x0D
Audio Mute Output Output 2	wM60002AU\x0D
Audio Mute Output Right Amplified Output	wM60009AU\x0D
Audio Mute Output Right Digital Output	wM60007AU\x0D
Audio Mute Output Right Variable Output	wM60005AU\x0D
Auto Switch Mode	wAUSW\x0D
Ducking Mic/Line 1	we44800au\x0D
Ducking Mic/Line 2	we44801au\x0D
Executive Mode	X
Freeze	F
Global Video Mute	B
Group Bass	wD5GRPM\x0D
Group Mic Mute	wD4GRPM\x0D
Group Mic Volume	wD3GRPM\x0D
Group Output Mute	wD7GRPM\x0D
Group Output Volume	wD8GRPM\x0D
Group Program Mute	wD2GRPM\x0D
Group Program Volume	wD1GRPM\x0D
Group Treble	wD6GRPM\x0D
HDCP Input Authorization Input 3	wE3HDCP\x0D
HDCP Input Authorization Input 6	wE6HDCP\x0D
HDCP Input Authorization Input 8	wE8HDCP\x0D
HDCP Input Status Input 3	wI3HDCP\x0D
HDCP Input Status Input 8	wI8HDCP\x0D
HDCP Output Status Output A	wO1HDCP\x0D
HDCP Output Status Output B	wO2HDCP\x0D
HDCP Output Status Output C	wO3HDCP\x0D
Input Signal Status Input 1	w0LS\x0D
Input Signal Status Input 6	w0LS\x0D
Input Signal Status Input 8	w0LS\x0D
Input Signal Type Input 1	1\
Input Signal Type Input 2	2\

Input Type Audio	$
Input Type Video	&
Line Input Gain Input 1 Type Analog	wG30000AU\x0D
Line Input Gain Input 1 Type LPCM-2Ch	wD30100AU\x0D
Line Input Gain Input 6 Type Analog	wG30010AU\x0D
Line Input Gain Input 6 Type LPCM-2Ch	wD30110AU\x0D
Line Input Gain Input 8 Type Analog	wG30014AU\x0D
Line Input Gain Input 8 Type LPCM-2Ch	wD30114AU\x0D
Mic Line Input Gain Input 1	wG40000AU\x0D
Mic Line Input Gain Input 2	wG40001AU\x0D
Mic Line Input Mute Input 1	wM40000AU\x0D
Mic Line Input Mute Input 2	wM40001AU\x0D
Mic Mix Level Mic 1	wG40100AU\x0D
Mic Mix Level Mic 2	wG40101AU\x0D
Mic Mix Mute Mic 1	wM40000AU\x0D
Mic Mix Mute Mic 2	wM40001AU\x0D
Output Format Output A	w1VTPO\x0D
Output Format Output B	w2VTPO\x0D
Output Format Output C	w3VTPO\x0D
Output Gain Output Amplified Output	wG60008AU\x0D
Output Gain Output Left Amplified Output	wG60008AU\x0D
Output Gain Output Left Digital Output	wG60006AU\x0D
Output Gain Output Left Variable Output	wG60004AU\x0D
Output Gain Output Output 1	wG60000AU\x0D
Output Gain Output Output 2	wG60002AU\x0D
Output Gain Output Right Amplified Output	wG60009AU\x0D
Output Gain Output Right Digital Output	wG60007AU\x0D
Output Gain Output Right Variable Output	wG60005AU\x0D
Output Resolution	wRATE\x0D
Power Save Mode	wPSAV\x0D
Screen Saver Status	wSSSAV\x0D
Switch Transition	wSWEF\x0D
Temperature	w20STAT\x0D
Test Pattern	wTEST\x0D
Video Mute Output A	1*B
Video Mute Output B	2*B
Video Mute Output C	3*B